

SupportOps Copilot

AI automation for MNC support teams: ticket triage, knowledge assistance, shift handover, and ServiceNow-driven server checks.

4**support workflows**

triage, KB, handover, SNOW automation

20+**synthetic tickets**

public-repo safe demo data

Solo builder: Kanhaiya Kumar Singh

Theme: AI at Work - Productivity & Teamwork Reimagined

SNOW -> AI -> Server checks -> Work notes

[INC0010422](#)**High memory on APP-SRV-014**

94% memory

[INC0010423](#)**Check SQL services**

SQLSERVERAGENT stopped

[TCK-1001](#)**403 after deployment**

Identity Platform

Support teams lose hours converting tickets into action.

MNC L1/L2 support work is still full of repetitive checks, copy-paste updates, and manual routing.

1**Manual triage**

Read SNOW, infer owner, assign group

2**Server checks**

RDP/remote shell, inspect memory and CPU

3**Service checks**

Find service names and verify running state

4**Ticket updates**

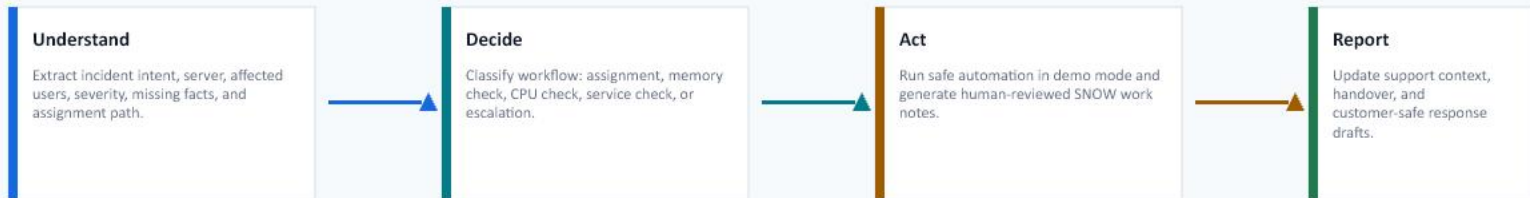
Write work notes, customer update, handover

The support engineer is not missing knowledge; they are trapped in repetitive coordination loops.

Impact: slower SLA response, inconsistent handovers, missed escalations, and higher fatigue during incident-heavy shifts.

SupportOps Copilot turns incident text into resolved next steps.

One dashboard coordinates AI understanding, SOP lookup, ticket updates, and safe automation recommendations.



The goal is not to replace support engineers. It removes the repetitive first 20 minutes so humans can handle judgment, approvals, and customer trust.

The prototype covers the real support shift loop.

From a messy incoming queue to analysis, runbook guidance, handover, and SNOW automation.

1 Ticket triage

Classify P1-P4, category, business impact, missing info, resolver group, and draft customer reply.

2 Knowledge assistant

Answers support questions from SOPs/runbooks and refuses unsupported guidance.

3 Shift handover

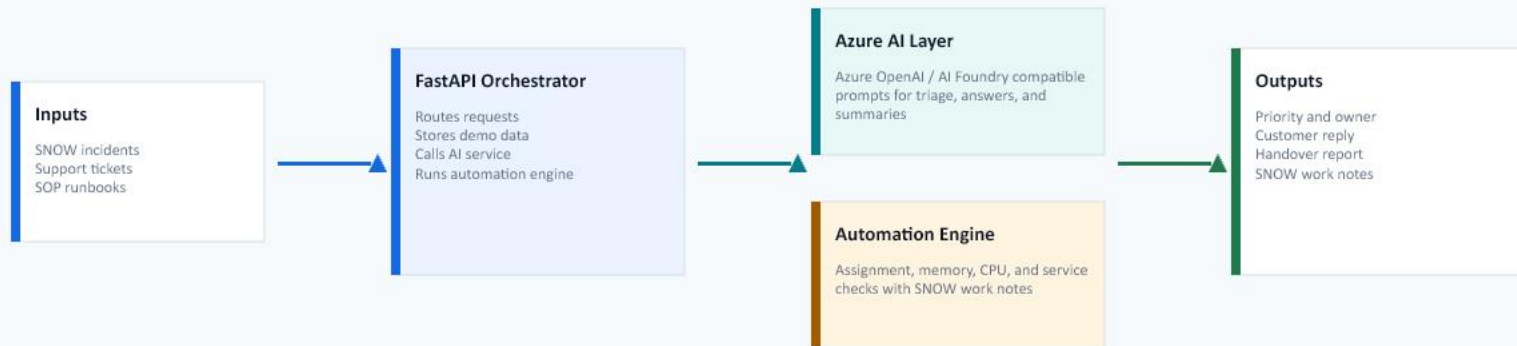
Groups open items into critical, blocked, waiting on customer, escalated, and next actions.

4 Version 2 SNOW automation

Simulates ServiceNow incident intake, assignment routing, memory check, CPU check, service status check, and work-notes draft.

A thin FastAPI layer orchestrates AI, SOPs, and automation.

The demo uses safe mock connectors now, with production paths for ServiceNow and server checks.



Production extension: ServiceNow Table API + WinRM/PowerShell remoting + Azure Key Vault secrets.

AI is used where language creates operational ambiguity.

The app keeps deterministic guardrails around every generated support action.

Triage prompt

Returns strict JSON: priority, category, business impact, affected users, missing info, next steps, escalation team, and reply.

Grounded SOP answer

Answers only from provided runbook context and says when information is missing.

Handover summary

Groups active tickets by risk, blocker, customer wait, engineering escalation, and next-shift actions.

Mock fallback

Keeps the prototype demoable when Azure credits or keys are delayed.

Human review

required on outputs

No autonomous customer-facing action

Strict JSON

for triage

Reduces hallucinated fields

Azure-ready

configuration

Endpoint, key, deployment via env vars

The working prototype shows the complete support path.

Dashboard frames mirror the live app flows that judges can interact with.

Ticket triage

P1 TCK-1001

APAC users cannot log in after deployment
Category: Authentication / Access
Escalation: Identity Platform
Next steps: check auth policy, SSO claims, rollback risk

Knowledge + handover

Question: What should I check for 403 after deployment?

Answer from SSO 403 runbook
Critical: all P1 tickets
Blocked: backup + incident bot
Next shift: update P1 every 30 min

V2 SNOW automation

INC0010422

High memory on APP-SRV-014
Memory: 94%
CPU: 38%
SNOW work notes drafted with next action and human approval warning

Live flow: analyze ticket -> ask SOPs -> generate handover -> run SNOW memory/service automation.

The value is fewer SLA misses and cleaner shift execution.

SupportOps Copilot targets the repetitive work that slows every incident-heavy support team.

Manual triage time

AI extracts priority, team, missing facts



80% automation potential

Handover gaps

Open risks grouped consistently



70% automation potential

Repeated SOP search

Runbooks answer common checks



65% automation potential

Ticket update effort

Work notes drafted from evidence



5% au

1 view

for shift context

Queue, AI, SOPs, automation

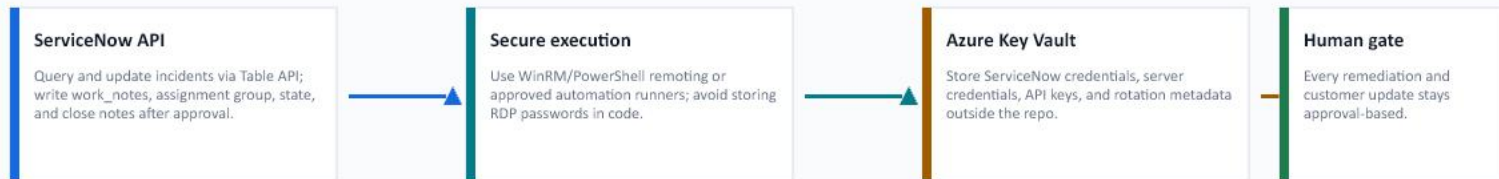
30 min

P1 update rhythm

Built into handover and customer notes

Production automation needs secure connectors, not hard-coded access.

The v2 prototype intentionally separates demo automation from real remediation.



Security posture: synthetic demo data, no secrets in GitHub, safe mock mode by default, and explicit human review warnings on outputs.

From copilot to governed support automation.

The next step is connecting real ITSM and infrastructure systems while preserving human approval.

1 ServiceNow live connector

Read incidents, classify use case, and update work notes through approved API credentials.

2 Server-check runner

Use secure PowerShell/WinRM jobs for memory, CPU, disk, and service health checks.

3 Approval workflow

Require engineer approval before service restart, reassignment, or customer-facing update.

4 Learning loop

Capture outcomes to improve routing and SOP recommendations.

SupportOps Copilot gives MNC support teams a practical path from AI assistance to trusted automation.

Repository: github.com/KanhaiyaSingh0/SupportOps-Copilot